

1. Administrative & Study Identification

Full Study Title: Phase 2b Multicentre, Randomized, Double-Blind, Placebo-Controlled, Parallel Group Dose-Ranging Study to Assess the Efficacy, Safety, and Tolerability of AZD2373 in Participants With APOL1-Mediated Kidney Disease **Short Title/Acronym:** APPRECIATE **Protocol Number:** D6800C00005 **NCT Number:** NCT06824987 **Sponsor Name:** AstraZeneca **Investigational Product:** AZD2373 (Opemalirsen) / Placebo **Phase of Development:** Phase 2b **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To assess the efficacy and safety of AZD2373 in participants diagnosed with APOL1-Mediated Kidney Disease (AMKD) who are homozygotes or compound heterozygotes for APOL1 high-risk genotypes (G1 and G2) by evaluating the relative change from Baseline in UACR at Week 30. **Secondary Objectives:** To evaluate the relative change in Urine Protein-Creatinine Ratio (UPCR) from Baseline at Week 30, evaluate pharmacokinetics, and assess the overall safety and tolerability profile.

2.2 Background & Rationale Patients with chronic kidney disease (CKD) who carry high-risk variants (G1, G2) of the APOL1 gene are at a significantly higher risk of progressive kidney disease and kidney failure. AZD2373 is an investigational antisense oligonucleotide designed to inhibit the production of the APOL1 protein. This study evaluates its potential to reduce proteinuria and slow disease progression compared to placebo.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** 1:1:1 (AZD2373 Arm 1, AZD2373 Arm 2, Placebo)

3.2 Planned Enrollment & Duration Target \$N\$: 96 participants (~32 per treatment group) **Number of Sites:** Multicenter (including US and UK) **Estimated Study Start:** 05-Mar-2025 **Estimated Primary Completion:** 30-Aug-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male and female participants aged 18 to 65 years, inclusive. 2. Confirmed high-risk APOL1 genotype (G1/G1; G1/G2; G2/G2). 3. A geometric mean UACR ≥ 300 mg/g and eGFR ≥ 25 mL/min/1.73m².

4.2 Exclusion Criteria 1. Participants on kidney replacement therapy (dialysis or kidney transplant) or any other organ transplant. 2. History or serologic evidence of autoimmune-mediated glomerular disease (e.g., lupus nephritis, IgA nephropathy). 3. Type 1 diabetes mellitus or poorly controlled Type 2 diabetes (e.g., past/current insulin use, HbA1c > 8.0%).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Subcutaneous injections of AZD2373 Dose A, AZD2373 Dose B, or Placebo. **Route of Administration:** Subcutaneous (SC) via Accessorized Pre-Filled Syringe. **Duration of Treatment:** Minimum of 30 weeks of study treatment.

5.2 Concomitant & Prohibited Therapy Permitted: Standard background medications for CKD and hypertension (e.g., SGLT inhibitors alongside up to one other oral anti-hyperglycaemic agent). **Prohibited:** New initiations of systemic immunosuppressive agents, insulin therapy, or other investigational therapies.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Up to Day 0	Informed Consent, APOL1 Genotyping Assay, baseline FMV urine samples for UACR, eGFR measurement
Baseline	Day 0	Randomization (1:1:1), First SC Injection, Vital Signs, Safety Labs
Interim	Weeks 1 to 29	Safety Labs, SC Dosing, Efficacy Assessment (UACR monitoring), PK profiling
End of Study	Week 30	Final UACR/UPCR Assessments, Exit procedures, transition into Open-Label Extension (OLE)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Greater reduction in Urine Albumin-Creatinine Ratio (UACR) as assessed by the relative change from Baseline in UACR at Week 30. **Measurement Method:** Central Laboratory Analysis calculating the geometric mean from 3 First Morning Void (FMV) urine samples collected on consecutive days.

7.2 Secondary & Exploratory Endpoints - Relative change in Urine Protein-Creatinine Ratio (UPCR) from Baseline at Week 30. - Safety, tolerability, and pharmacokinetic (PK) parameters.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring at study visits, including assessment of injection site reactions and renal function. **Serious Adverse Events (SAEs):** Standard 24-hour reporting timeframe for severe, life-threatening, or hospitalization-inducing events. **Data Monitoring Committee (DMC):** Independent safety review board to monitor blinded data.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy; Safety Set for all participants receiving at least one dose of the investigational product. **Sample Size Justification:** $N=96$ (~32 per arm) powered to detect a dose-response effect of AZD2373 vs. placebo on the percentage change in UACR. **Handling of Missing Data:** Appropriate imputation methods (e.g., Mixed Models for Repeated Measures) for early withdrawals prior to Week 30.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine onsite and remote monitoring to ensure protocol adherence, accurate central lab testing, and appropriate storage of SC injections. **Audit Trail:** Secure electronic Case Report Form (eCRF) utilizing verified audit logs for all clinical and laboratory data entries.

11. Study Identifiers & Governance

Primary ID: {{D6800C00005}} **Registry Number:** {{NCT06824987}}
Sponsor/Lead Organization: {{AstraZeneca}} **Study Title:** {{APPRECIATE}} **Official Regulatory Title:** {{Phase 2b Multicentre, Randomized, Double-Blind, Placebo-Controlled, Parallel Group Dose-Ranging Study to Assess the Efficacy, Safety, and Tolerability of AZD2373 in Participants With APOL1-Mediated Kidney Disease}}

12. Clinical Framework (AMKD Specific)

Primary Condition: {{APOL1-Mediated Kidney Disease (AMKD)}}
Diagnosis Threshold: {{High-risk APOL1 Genotype (G1/G1; G1/G2;}}

G2/G2) & UACR $\geq$ 300 mg/g}} **Medical Methodology:** {{Antisense Oligonucleotide (ASO) RNA Therapy}} **Optimization Target:** {{Reduction of UACR at Week 30}}

13. Study Procedures & Methodology

Management Pathway: {{Diagnostic APOL1 Genotyping Clinical Trial Assay followed by targeted SC ASO therapy}} **Education Protocol (HCP):** {{Investigator training on SC pre-filled syringe administration and FMV sample acquisition}} **Education Protocol (Patient):** {{Informed consent regarding genetic testing parameters and adherence to daily FMV urine collections}} **Quality Audit Mechanism:** {{Centralized genetic assay verification and blinded placebo-controlled data management}}

14. Observation & Follow-Up Schedule

Total Duration: {{30 Weeks of Double-Blind Treatment}} **Onsite Visit Cadence:** {{Screening, Baseline, and regular dosing intervals up to Week 30}} **Remote Monitoring Frequency:** {{As needed for safety check-ins between dosing visits}} **Checklist Review Interval:** {{Prior to each SC dose administration to assess AE/SAE development}}

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	03-Apr-2026				
Clinical Operations Lead	[Name]	____	03-Apr-2026		Lead		
Statistician	[Name]	____	03-Apr-2026				